

CDS DS 120: Foundations of Data Science

We love Maths!!!!

Instructor Name: Dr.Tanima Chatterjee

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Office Hours: Monday: 12:15pm-1:00pm, Wednesday: 12:15pm-1:00pm, Friday: 12:15pm-1:00pm in CCDS 1402

TA1: Sahana Kowshik

OH: Monday: 10:00am – 11:00am and Friday: 10:00am-11:00am, in yellow corner

on 14th Floor near CCDS 1411.

Discussion: A4, A3, B3

Email: skowshik@bu.edu

TA2: Srushti Karve

OH: Friday: 2:30pm-4:00pm, Wednesday: 1:00pm-2:00pm in yellow corner on 14th Floor near CCDS 1411.

Discussion: A5, B5

Email: srushtik@bu.edu

TA3: Sungjoon Park

OH: Tuesday: 1:30pm - 3:30pm in yellow corner on 14th Floor near CCDS 1411.

Discussion: A2, B2, B4

Email: spark618@bu.edu

CA1: Zihao Guo

OF: Tuesday: 2:00pm – 3:30pm in yellow corner on 14th Floor near CCDS 1411.

Email: z2004@bu.edu

CA2: Riwaz Shrestha

OF: Monday: 3:30pm – 5:00pm in yellow corner on 14th Floor near CCDS 1411.

Email: riwaz@bu.edu

CA3: Hassan Dawy

OF: Thursday: 10:30am- 12:00pm in yellow corner on 14th Floor near CCDS 1411.

Email: hdawy@bu.edu

CA4: Celia Gao

OF: Wednesday: 2:30pm- 4:00pm in yellow corner on 14th Floor near CCDS 1411.

Email: celiag@bu.edu

Course Description

CDS DS 120 is the first in a three-course sequence (including CDS DS 121 and CDS DS 122) that introduces students to theoretical foundations of Data Science. The three courses take an iterative (spiral) approach to a set of foundational material in Mathematical Logic, Calculus, Linear Algebra, and Probability and more.

DS 120 covers an introduction to key concepts from Proofs, Logic, Linear Algebra (vector spaces, matrices, and linear systems), Calculus (Differentiation and

Integration), Probability (discrete and continuous random variables) and introduction to Statistics. These areas find culminating applications in Algorithms, hypothesis testing, gradient descent for neural networks, and regression. A detailed topic list is provided at the end of this syllabus.

The course is aimed at building the foundations of mathematical thinking, specifically through the use of problem sets that require students to answer mathematically-posed questions.

Hub Learning Outcomes

Quantitative Reasoning I (QR1) Learning Outcomes:

Students will develop understanding of a large set of fundamental theoretical tools used in quantitative reasoning: Mathematical logic, Induction, Calculus, Random variables, Probability distributions, Summary statistics, Vector Spaces, Matrices, and Linear systems. Through problem solving during discussion sections, students will develop skills in using computation to answer questions posed in mathematical terms.

Students will learn to interpret data in terms of probability distributions and also in terms of vectors. Students will develop skills presenting data visually in terms of plots, histograms, etc; symbolically in terms of matrix-vector equations and linear systems; and numerically in terms of summary statistics and fitted model parameters.

Other Learning Outcomes (e.g., School, Department, and/or Program

Outcomes) On completion of DS 120, students will achieve proficiency in:

- Using probability to describe data and construct models, including models involving univariate distributions and discrete and continuous random variables.
- Using basic linear algebra (vectors and matrices) to describe and solve

systems of linear equations, to represent data as points in vector spaces, to represent linear transformations in terms of matrices and matrix inverses, and to represent distances and lengths in terms of norms and inner products.

- Deriving derivatives of functions with multiple variables, and using differentiation to solve problems.
- Understanding mathematical statements and logic, thereby gaining proficiency in the language of mathematics, along with specific proof techniques.

This course contributes to the learning outcomes of the BS in Data Science with respect to:

- Foundations:
 - Mastery of the principal tools of data driven decision-making, including defining models, learning model parameters, and making predictions.
 - Understanding of how to use probabilistic models to make decisions.
- Integration and Application:
 - Ability to interpret and explain results .

Instructional Format, Course Pedagogy, and Approach to Learning

This course will be primarily a lecture heavy course where students will learn the mathematical foundations required for interpreting data during the lectures. The lectures don't require students to carry laptops. It will be pen and paper lectures with in class participation. The discussion sessions will concentrate on applying the concepts learnt during the lectures to solve mathematical questions along with a review of the topics learnt during the lecture and on practicing further problems to bolster students' understanding of the concepts. The discussions will also have

surprise *pop quizzes* to gauge students' understanding and engagement. The HomeWorks will serve as a method of applying the concepts learnt during the lectures and discussions to applied problems. Homeworks will be released on a weekly basis on Fridays and students will be given a week to solve the problems. HomeWorks are also meant to check the students understanding of the concepts taught in class.

Books and Other Course Materials

The suggested textbook (not compulsory) for this class can be purchased from BU bookstore or from the following link:

<https://he.kendallhunt.com/product/mathematics-data-science>

The students may find the following free references useful for the course material: Hefferon, Linear Algebra. Available free online:

<http://joshua.smcvt.edu/linearalgebra/>

Lane, Online Statistics Education: A Multimedia Course of Study. Available free online:

<http://onlinestatbook.com/>

Strang, Calculus. Available free online:

<https://ocw.mit.edu/ans7870/resources/Strang/Edited/Calculus/Calculus.pdf>

A guided journey into the world of abstract mathematics and the writing of proofs:

https://www.math.cmu.edu/~jmackey/151_128/bws_book.pdf

Courseware

We'll be using **Blackboard** and **Gradescope** for distributing assignments, collecting assignments, grading, and the distribution of course notes:

<https://learn.bu.edu>

We'll be using **Piazza** for Q&A.

Gradescope entry code: X8EYXZ (For Section A)

G6753E (For Section B)

Piazza Sign up link: <https://piazza.com/bu/spring2026/ds120>

Assignments and Grading

Students will be tested via *two midterms* and a *final exam*. All of these are written, in-class exams.

Students would also be evaluated on their *HomeWorks* which would be released on *Gradescope* on a weekly basis *every Friday*. The HomeWork assignments are meant to be relatively short with occasional challenge questions to ensure the students' understanding of the concepts taught in class. HomeWorks will be due by the end of Friday of the next week. **No HomeWorks sent via emails will be entertained. THERE WILL BE NO EXCEPTIONS TO THIS RULE.**

There will be occasional *surprise pop quizzes* during the *discussions*. These will account for an *additional 5%* of the students' grades which will be automatically added to the students' final cumulative grades (capping at a total cumulative final score of 100%).

Student *participation* in class (**especially discussions**) along with their involvement in Piazza will hold value as well.

If a student fails to attend at least 60% of the Discussions, they would not be graded for their Pop Quizzes.

Grading:

15% Homework Assignments

35% Final Exam

20% Midterm1

20% Midterm2

10% Class Participation (Lecture, Piazza and **Discussion**)

Bonus (extra) 5% **Discussion Pop Quizzes**

The grade cutoffs will be determined after all the exams and assignments are graded and will depend on the difficulty of the exams/assignments and overall class performance as determined by the instructor. The grade boundaries will not be disclosed.

*** While the graded midterms will be available to the students for review on Gradescope after grading, the final exams will not be available to the students for review. The students can, however, email the instructor within 48 hours after the final exam scores are released if they feel there is a grading error. ***

Resources/Support/How to Succeed in This Course:

1. To succeed in this course students should attend all lectures, come to discussion prepared with questions, complete all assignments on time, and discuss problems and material with fellow classmates.
2. Students are welcomed and encouraged to visit office hours.
3. 4. The [Education Resource Center](#) offers free individual and group tutoring.

Accommodations for Students with Documented Disabilities: If you are a student with a disability or believe you might have a disability that requires accommodations, please contact the Office for Disability Services (ODS) at (617) 353-3658 or access@bu.edu to coordinate any reasonable accommodation requests. ODS is located at 25 Buick Street on the 3rd floor.

Community of Learning: Class and University Policies

1. **Courtesy expectations.** Students are responsible for supporting a courteous learning environment. Please show respect for other students' questions, and maintain an attentive attitude in class. If you have an advanced question to ask, try to make it accessible to the rest of the class. Explain things to other students with a

polite tone.

2. Attendance & Absences. Attending class is optional but highly encouraged; behavior in class does carry some small participation weight. ***Regular absence without notice will, however, be penalized.*** Attending recitation carries some participation credit. However, either can be waived with a reasonable request for accommodation. We will in particular waive for religious holidays; see the University [Policy on Religious Observance](#).

3. Assignment Completion & Late Work. Assignments will be submitted as PDF on Gradescope. After the deadline, **NO** late submissions will be allowed. ***NO HomeWorks submitted via email will be graded.***

4. Academic Conduct Statement

Students are expected to abide by the guidelines and rules of the Academic Code of Conduct. <https://www.bu.edu/academics/policies/academic-conduct-code/>

5. Using Generative AI in coursework. The students are expected to abide by the guidelines of the GenAI Policy as given in: <https://www.bu.edu/cds-faculty/culture-community/gaia-policy/>

General Course Plan with Topics to be Covered: (Not written in stone!)

HomeWorks will be due by Friday 11:59 pm of the week following the week the HomeWork was released.

Module 1: Truths and Proofs

Introduction to Logic: Understanding mathematical statements and truth values, Existential vs. Universal Quantifiers, Specifying Quantification sets

Negation of Mathematical statements

Conditional statements, truth tables, Set Operations and Logical Connectives,
Converse and Contrapositives, Biconditional Statements

Necessary and Sufficient Conditions, Proving logical equivalences

Introduction to proofs by Mathematical Induction

Module 2: Linear Algebra

Introduction to Linear Algebra, Gaussian elimination, RREF

Introduction to vectors and Matrices, Operations on Vectors and Matrices

Introduction to Vector spaces, Linear independence and Bases

Ranks and determinants

Projections

Linear Regression: Line of best fit, More linear fits, Introduction to Multiple Regression

Module 3 : Calculus and Gradient Descent

Introduction to Multivariable Calculus:

Introduction to Gradient: Second order Partial Derivative test, Directional Derivatives

Introduction to Gradient Descent and its limitations

Review of Integral Calculus

Module 4: Probability and Statistics

Introduction to Combinatorics

Probability

Conditional Probability, Joint Distribution, Introduction to Bayes Rule

Expectation

Probability Distribution and variance

Gaussian Distribution and Central Limit Theorem

Confidence Intervals and t-Distributions

Important Dates (Tentative! Mark the week of the exams in your calendars)

Midterm1: February 23rd, 2026 (Monday)

Midterm2: March 27th, 2026 (Friday)

Final: MARK YOUR CALENDARS! TBD by Registrar's Office (Week of May 4th -May 8th)